

Tentacles Fever: Sound-Responsive SMA Actuators

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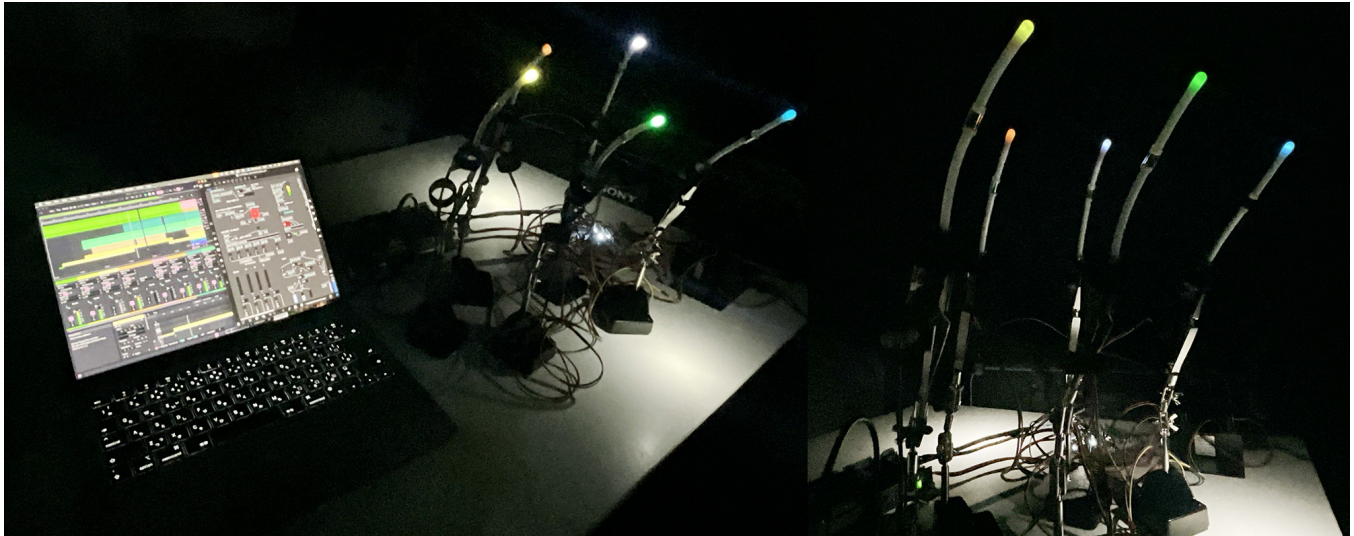


Figure 1: Snapshots of Tentacle Fever.

Abstract

This work aims to visualize sound and music through the movement of tentacle-like actuators. The actuator bends smoothly and is driven by BioMetal fiber, a shape-memory alloy wire. Each actuator contains four BioMetal fibers, enabling it to bend in four directions. By synchronizing their movement with sound, we seek to create an embodied expression reminiscent of a living creature dancing to music.

CCS Concepts

• **Hardware** → **Sensors and actuators.**

Keywords

Sound Responsive, Shape Changing Interface, Soft Actuator, Shape-Memory Alloy

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1 Introduction

There are many techniques for visually expressing sound and music, ranging from methods that move or transform physical objects to software-based sound visualizers. Examples of physical phenomena include magnetic fluids [Kodama and Miyajima 2007] and flames [Fysikshow 2014], and some works integrate computer technology with cymatics [Stanford 2014], a phenomenon in which sound waves and vibrations create geometric patterns in materials like particles or liquids. Music-synchronized fountains and fireworks are also common examples.

Humanoid robots [Dynamics 2020] and anthropomorphic toys [Tomy Co. 1988] are not merely visualizations but performative expressions that appear to dance to music. In sound sculptures [Lieberman and Paluska 2008; Zimoun 2013] where machines generate sound or perform music, the mechanisms and movements themselves form an integral part of the artistic expression and contribute to the visualization of sound. Robotic performances [Weinberg 2015], like those by humans, are also visual experiences, with the robots' movements enhancing and complementing the accompanying sound and music.

These sound-related expressions can be found across various domains, including products, toys, artworks, and lighting, using a wide range of materials and media. This diversity shows that the visualization of sound resonates with and attracts many people.

We have conducted a series of experiments and developed interactive artworks [Nakayasu 2020, 2024] using a tentacle-like actuator capable of bending in multiple directions. In this work, we aim to

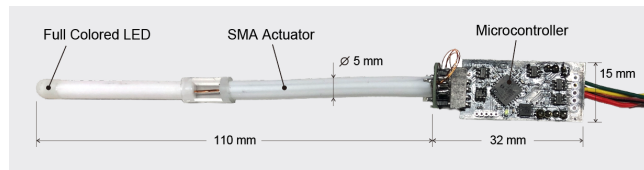


Figure 2: Actuator unit

visualize sound and music through the dynamic motion of tentacle-like actuators. By synchronizing their movement with sound, we seek to create an embodied expression reminiscent of a living creature dancing to music.

2 Implementation

2.1 System Overview

The Tentacles Fever system consists of a project created in Ableton Live, a control program developed with Cycling '74 Max, and an actuator unit. The Ableton Live project is composed of 20 individual instrument tracks. Since this version of Tentacles Fever uses five actuators, the original tracks are reorganized into five composite tracks based on their musical characteristics. By switching instruments according to the musical progression, the system enables the visualization of music without losing its expressive qualities, even with a limited number of actuators.

2.2 Design of the Actuator Unit

The actuator unit includes an SMA actuator and a microcontroller board (Fig. 2). The SMA actuator consists of a silicone tube (5 mm in diameter and 110 mm in length), four BioMetal fibers (thin SMA wires developed by TOKI Corporation), a full-color LED chip, and a diffuser cap. The four BioMetal fibers drive the actuator, allowing it to bend in four directions. Each fiber contracts when heated and returns to its original length upon cooling. Heating is controlled using pulse-width modulation (PWM) currents, which are generated by a control circuit comprising a peripheral interface controller (PIC), a field-effect transistor (FET), and other electronic components. The actuator unit is operated via DMX signals.

3 Actuator Behavior in Response to Sound

In this system, rhythm is visualized through the oscillatory motion of actuators. Humans tend to perceive rhythm not at the initiation of motion but at its return point. As a result, during real-time playback, the actuator's motion may appear delayed relative to the music. To address this, two identical audio tracks are played simultaneously: one for rhythm detection and the other for auditory output, delayed by approximately 1.5 seconds. This temporal offset allows the actuator's motion to appear more rhythmically synchronized with the music.

This configuration creates the impression that the tentacles are physically interacting with the sound—almost as if conducting the music. The bending direction of each actuator is randomly switched to produce motion that avoids mechanical repetition. Occasionally, all five actuators bend in the same direction by chance, resulting in a unison-like movement that evokes a sense of harmony and satisfaction in viewers. This balance between stochastic variation and

emergent synchrony contributes to an expressive visual behavior that appears to dance with the music, drawing the viewer in.

In this implementation, five actuators are used, each corresponding to a group of instrument tracks. These groupings are based on musical characteristics, allowing each actuator to reflect the rhythm and texture of different instrumental elements within the composition. This approach enables a meaningful level of musical expression, even with a limited number of actuators.

With a larger number of actuators, it would be possible to assign individual actuators to specific instruments or even to particular musical notes. Ultimately, a high-density actuator array could serve as a highly expressive medium for visualizing sound waves in complex spatial patterns, akin to cymatics. Such a system has the potential to greatly enhance the expressive capabilities of physical sound visualization in interactive and performative contexts.

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